

# GOOD SOIL STEM ACADEMY — MASTER KIT SPECIFICATION & PROCUREMENT TABLE

Complete 25-Class Kit Inventory for Junior Batch (Ages 6–9 / Grades 1–4) & Senior Batch (Ages 10–14 / Grades 5–9)

| Program Metric   | Junior Batch (Ages 6–9 / Grades 1–4)                          | Senior Batch (Ages 10–14 / Grades 5–9)                           | Procurement & Logistics Architecture                         |
|------------------|---------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|
| Curriculum Scope | 25 Hands-on Classes (50 Hours instruction)                    | 25 Hands-on Classes (50 Hours instruction)                       | Weekly 2-hour weekend modules across 6 months                |
| Kit Distribution | 10 Take-Home Kits + 10 Group Sets + 4 Challenges + 1 Expo     | 10 Take-Home Kits + 10 Group Sets + 4 Challenges + 1 Expo        | Take-home kits retained by students; Studio kits reused      |
| Domain Coverage  | Aerodynamics, Basic Circuits, Gears, Bionics, Optics, Science | Hydraulics, Advanced Aero, Electronics, Sensors, Autonomous Bots | 7 Verified Direct Vendors (Pludo, Mindtronix, Kiddale, etc.) |

## 1. Junior Batch (Ages 6–9 / Grades 1–4) — Complete 25-Class Kit List

Focus: Tactile physical mechanics, basic circuits, introductory aerodynamics, optics, chemical reactions, and bionic linkages. (10 Take-Home Giveaway Kits + 10 Group Studio Kits + 4 Milestones + 1 Expo)

| Cls | Kit / Activity Name                     | Domain         | Format / Type | Vendor              | SKU / Code       | Unit Price | Core Learning Concept & Key Pedagogical Outcome                                             |
|-----|-----------------------------------------|----------------|---------------|---------------------|------------------|------------|---------------------------------------------------------------------------------------------|
| 01  | Electric Propeller & Wind Car           | Electricity I  | Take-Home #1  | Pludo (Robocraze)   | WEB-PLU-015      | Rs. 467    | Basic DC motor circuits, propeller thrust, aerodynamics, switch control.                    |
| 02  | High-Lift Paper Gliders & Launcher      | Aero I         | Group Kit #1  | Good Soil Aero      | CUSTOM-AERO-01   | Rs. 150    | Airfoil cross-sections, lift vs gravity, thrust launcher, wing aspect ratio.                |
| 03  | Kitchen Chemistry & Color Fireworks     | Science I      | Group Kit #2  | Butterfly Edufields | WEB-BUT-051      | Rs. 649    | Acid-base neutralization, CO2 gas generation, liquid density layers, color kinetics.        |
| 04  | Wooden Catapult & Trajectory Shooter    | Mechanics I    | Take-Home #2  | Practiko            | WEB-PRA-052      | Rs. 499    | Class 3 levers, elastic potential energy, fulcrum position, trajectory angles.              |
| 05  | Spin Art & Craft STEAM Machine          | STEAM Art I    | Take-Home #3  | Butterfly Edufields | WEB-BUT-060      | Rs. 399    | Centrifugal force, rotational dynamics, symmetry, electric motor speed blending.            |
| 06  | Assessment 1: Island Escape Challenge   | Challenge 1    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-01 | Rs. 200    | Flotation engineering challenge & catapult target accuracy quest (Badge: Explorer).         |
| 07  | Wooden Paddle Steamer Propeller Boat    | Mechanics II   | Take-Home #4  | Kiddale             | WEB-KDA-031      | Rs. 424    | Water displacement, paddle propulsion, buoyant force, rotary energy conversion.             |
| 08  | Balsa/Foam Chuck Glider & Flight Tuning | Aero II        | Group Kit #3  | Good Soil Aero      | CUSTOM-AERO-02   | Rs. 200    | Center of Gravity (CG) balancing, elevator/rudder trim tabs, dihedral stability.            |
| 09  | Wooden Glow Table Lamp & Switch Circuit | Electricity II | Take-Home #5  | Kiddale             | WEB-KDA-042      | Rs. 424    | Closed vs open loops, LED anode/cathode polarity, conductor continuity, woodcraft assembly. |
| 10  | Wooden Mechanical Carousel & Gear Train | Mechanics III  | Group Kit #4  | Kiddale             | WEB-KDA-032      | Rs. 424    | Compound spur gear trains, speed reduction vs torque increase, bevel gear direction.        |
| 11  | Morse Code Telegraph & Buzzer Station   | Electronics I  | Take-Home #6  | Studio / DIY        | STUDIO-ELEC-01   | Rs. 150    | Electromagnetic buzzers, momentary push switches, Morse code encoding & decoding.           |
| 12  | Assessment 2: STEAM Carnival Ride       | Challenge 2    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-02 | Rs. 200    | Multi-gear motorized carnival ride build & team showcase (Badge: Master Crafter).           |
| 13  | Pneumatic Air-Powered Rocket Launcher   | Aero III       | Take-Home #7  | Pludo (Robocraze)   | WEB-PLU-014      | Rs. 370    | Pneumatic pressure accumulation, sudden release valve, rocket fin stabilization.            |

| CI's | Kit / Activity Name                   | Domain         | Format / Type | Vendor              | SKU / Code       | Unit Price | Core Learning Concept & Key Pedagogical Outcome                                               |
|------|---------------------------------------|----------------|---------------|---------------------|------------------|------------|-----------------------------------------------------------------------------------------------|
| 14   | 10-in-1 Ultimate Electricity Lab      | Electronics II | Group Kit #5  | Butterfly Edufields | WEB-BUT-030      | Rs. 589    | Series vs parallel connections, magnetic reed switches, conductor vs insulator testing.       |
| 15   | Mechanical Secret Cipher Wheel & Maze | Coding I       | Take-Home #8  | Practiko            | WEB-PRA-068      | Rs. 499    | Caesar shift cipher, substitution algorithms, binary secret encoding, logical decision trees. |
| 16   | 3-Ball Motorized Sun, Earth & Moon    | Space Science  | Group Kit #6  | Kiddale             | WEB-KDA-139      | Rs. 449    | Orbital mechanics, solar/lunar eclipses, diurnal rotation, seasonal axial tilt.               |
| 17   | Pin Hole Camera & Optical Periscope   | Science II     | Take-Home #9  | Practiko            | WEB-PRA-054      | Rs. 499    | Rectilinear propagation of light, focal inverted projection, 45-degree mirror reflection.     |
| 18   | Assessment 3: Sky & Signals Derby     | Challenge 3    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-03 | Rs. 200    | Precision rocket range trials + optical mirror signal decoding race (Badge: Flight Cadet).    |
| 19   | Wooden Walking T-Rex Bionic Robot     | Robotics I     | Group Kit #7  | Kiddale             | WEB-KDA-039      | Rs. 594    | Four-bar linkage kinematics, crank-rocker motion, biomimetic 2-legged walking balance.        |
| 20   | Electric Coin-Eating Smart Bank Robot | Robotics II    | Group Kit #8  | Kiddale             | WEB-KDA-133      | Rs. 599    | Automated mechanical switch triggers, dual-jaw linkage, motor stall torque.                   |
| 21   | Wooden Solar Power Rover Car          | Clean Energy   | Take-Home #10 | Kiddale             | WEB-KDA-043      | Rs. 594    | Photovoltaic energy conversion, solar panel angle tuning, direct motor drive efficiency.      |
| 22   | Two-Wheeled Obstacle Rover Team Build | Robotics III   | Group Kit #9  | Kiddale             | WEB-KDA-140      | Rs. 499    | Differential drive steering, dual-motor synchronization, chassis center of gravity.           |
| 23   | Mini Quad Drone Discovery & Pilot     | Drones         | Group Kit #10 | BrainHap / Studio   | WEB-BRH-003      | Rs. 2,999  | Quad-rotor aerodynamics, roll/pitch/yaw control, gyro stabilization, pilot flight arena.      |
| 24   | Assessment 4: Future City Prototype   | Challenge 4    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-04 | Rs. 250    | Capstone integration: Combine bionic mechanisms + solar + circuits (Badge: Young Innovator).  |
| 25   | Grand Expo: Medals & Certs            | Graduation     | Grand Expo    | Good Soil Studio    | STUDIO-EXPO-01   | Rs. 300    | Live flight arena, robot racing, parent demo day, STEM Passport stamps & medals.              |

2. Senior Batch (Ages 10–14 / Grades 5–9) — Complete 25-Class Kit List

**Focus:** Advanced fluid hydraulics, projectile physics, sensor integration (acoustic/strain/ultrasonic), motor polarity logic, bionic multi-leg locomotion, and drone avionics. *(10 Take-Home Giveaway Kits + 10 Group Studio Kits + 4 Milestones + 1 Expo)*

| Cls | Kit / Activity Name                      | Domain         | Format / Type | Vendor              | SKU / Code        | Unit Price | Core Learning Concept & Key Pedagogical Outcome                                                             |
|-----|------------------------------------------|----------------|---------------|---------------------|-------------------|------------|-------------------------------------------------------------------------------------------------------------|
| 01  | Hydraulic Mech Gripper Robot Arm         | Mechanics I    | Take-Home #1  | Kintaro DIY         | WEB-KIN-001       | Rs. 749    | Pascal's principle, fluid mechanical advantage, master/slave hydraulic cylinders, 2-DOF gripper.            |
| 02  | Rubber-Band High-Endurance Monoplane     | Aero I         | Group Kit #1  | Good Soil Aero      | CUSTOM-AERO-03    | Rs. 400    | Cambered airfoil aerodynamics, rubber motor torque curve, propeller pitch optimization.                     |
| 03  | 150+ Chemistry & Physics Lab             | Science I      | Group Kit #2  | Butterfly Edufields | WEB-BUT-052       | Rs. 669    | Exothermic & endothermic reactions, universal pH titration, precipitation, crystal nucleation.              |
| 04  | Precision Force Vector Catapult          | Mechanics II   | Take-Home #2  | Mindtronix          | WEB-MIN-003       | Rs. 712    | Parabolic kinematics, projectile velocity vectors, adjustable tension stop angles, target math.             |
| 05  | SolarGlide Kinetic Solar Aircraft        | STEAM Art I    | Take-Home #3  | Pludo (Robocraze)   | WEB-PLU-024       | Rs. 719    | Rotary solar counterweight balance, low-friction bearing pivot, kinetic shadow aerodynamics.                |
| 06  | Assessment 1: Medieval Siege Challenge   | Challenge 1    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-01S | Rs. 250    | Vector force target competition + chemical indicator identification sprint (Badge: Apprentice Engineer).    |
| 07  | DIY Electronic Weighing Scale            | Electronics I  | Take-Home #4  | Mindtronix          | WEB-MIN-017       | Rs. 623    | Piezoresistive strain gauge deflection, Wheatstone bridge concept, tare calibration logic.                  |
| 08  | Large Wingspan Foam Glider & CG Trim     | Aero II        | Group Kit #3  | Good Soil Aero      | CUSTOM-AERO-04    | Rs. 500    | Glide ratio (L/D) maximization, aerodynamic stall mitigation, lateral dihedral stability.                   |
| 09  | 2-Way Reversible Motor Electric Car      | Electronics II | Take-Home #5  | Pludo (Robocraze)   | WEB-PLU-064       | Rs. 710    | DPDT rocker switch polarity inversion, H-bridge fundamentals, drivetrain traction tuning.                   |
| 10  | Heavy-Duty Cable Car Overhead Lift       | Mechanics III  | Group Kit #4  | Mindtronix          | WEB-MIN-015       | Rs. 759    | Compound pulley systems, mechanical tension vectors, high-torque reduction gearbox.                         |
| 11  | Neon Glow Optical Infinity Mirror        | STEAM Art II   | Take-Home #6  | Studio / DIY        | STUDIO-ART-01S    | Rs. 350    | Two-way semi-reflective mirrors, recursive light ray geometry, LED perimeter array.                         |
| 12  | Assessment 2: Rescue Rover Canyon Race   | Challenge 2    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-02S | Rs. 250    | Cable-driven payload transfer across obstacle gorge & reverse polarity time trial (Badge: Systems Builder). |
| 13  | Motorized Electric DIY Aircraft          | Aero III       | Take-Home #7  | Pludo (Robocraze)   | WEB-PLU-074       | Rs. 629    | High-RPM coreless motor thrust, lithium capacitor discharge rate, powered climb profiles.                   |
| 14  | Motorized Forklift & Heavy Cargo Lifter  | Mechanics IV   | Group Kit #5  | Mindtronix          | WEB-MIN-007       | Rs. 702    | Self-locking worm drive gearboxes, vertical rack-and-pinion carriage, load center stability.                |
| 15  | Clap-Racer Sound-Activated Car           | Sensors I      | Take-Home #8  | Pludo (Robocraze)   | WEB-PLU-009       | Rs. 719    | Electret microphone transducer, transistor trigger latch, acoustic noise threshold trimming.                |
| 16  | Orbital Satellite & Telemetry Antenna    | Space Tech     | Group Kit #6  | Pludo (Robocraze)   | WEB-PLU-019       | Rs. 629    | Geostationary vs LEO satellite orbits, solar panel orientation tracking, signal coverage cones.             |
| 17  | Infinity Curve & Harmonic Pendulum       | Science II     | Take-Home #9  | Kintaro DIY         | WEB-KIN-014       | Rs. 799    | Coupled harmonic oscillators, phase relationship equations, resonance damping, Lissajous curves.            |
| 18  | Assessment 3: Flight Derby & Sensor Maze | Challenge 3    | Assessment    | Good Soil Studio    | STUDIO-ASSESS-03S | Rs. 250    | Powered aircraft duration trial + sound-triggered obstacle navigation course (Badge: Avionics Specialist).  |
| 19  | Multi-Axis Motorized Robot Claw Machine  | Robotics I     | Group Kit #7  | Pludo (Robocraze)   | WEB-PLU-075       | Rs. 629    | 3-axis Cartesian mechanical linkages, bidirectional motor control, end-effector slip friction.              |

| Cls | Kit / Activity Name                   | Domain         | Format / Type | Vendor           | SKU / Code        | Unit Price | Core Learning Concept & Key Pedagogical Outcome                                                            |
|-----|---------------------------------------|----------------|---------------|------------------|-------------------|------------|------------------------------------------------------------------------------------------------------------|
| 20  | Bujji Autonomous Obstacle Robot       | Robotics II    | Group Kit #8  | Mindtronix       | WEB-MIN-022       | Rs. 799    | Ultrasonic time-of-flight echo sensing, obstacle avoidance decision state machine.                         |
| 21  | Programmable Motorized Arrow Launcher | Logic & Coding | Take-Home #10 | Mindtronix       | WEB-MIN-006       | Rs. 786    | Cam-driven rapid fire release, microswitch sequence timing, mechanical automation safety.                  |
| 22  | Moto-Bot Bionic Multi-Leg Walker      | Robotics III   | Group Kit #9  | Kintaro DIY      | WEB-KIN-015       | Rs. 799    | Theo Jansen & Klann linkage gait synthesis, multi-leg coordinate phase shifting, terrain traversal.        |
| 23  | BrainHap HD Camera Drone Piloting     | Drones         | Group Kit #10 | BrainHap         | WEB-BRH-003       | Rs. 2,999  | 6-axis gyroscope flight trim, optical flow altitude hold, FPV telemetry live stream diagnostics.           |
| 24  | Assessment 4: Shark Tank Defense      | Challenge 4    | Assessment    | Good Soil Studio | STUDIO-ASSESS-04S | Rs. 300    | Engineering prototype defense, circuit logic explanation, live performance demo (Badge: Master Innovator). |
| 25  | Senior Tech Symposium & Expo          | Graduation     | Grand Expo    | Good Soil Studio | STUDIO-EXPO-02    | Rs. 350    | Public technical symposium, autonomous robot battle arena, flight exhibition, trophies.                    |

### 3. Optional Supplementary Kits for Early Learners (Ages 3–6 / Preschool)

Modular early-childhood activity sets available for introductory weekend workshops or sibling daycare engagement.

| # | Kit / Resource Name                  | Age Group | Domain             | Vendor              | SKU Code    | Unit Price | Kit Description & Educational Focus                                              |
|---|--------------------------------------|-----------|--------------------|---------------------|-------------|------------|----------------------------------------------------------------------------------|
| 1 | 155 Alphabets & Words Magnet Lab     | 3–6 yrs   | Literacy & Phonics | Butterfly Edufields | WEB-BUT-063 | Rs. 549    | Soft EVA foam magnetic letter blocks, spelling flashcards, phonics word builder. |
| 2 | 90+ pcs Magnetic Numbers & Shapes    | 3–5 yrs   | Early Math & Logic | Butterfly Edufields | WEB-BUT-061 | Rs. 499    | Geometric shape recognition, tactile number puzzles, early addition/subtraction. |
| 3 | Air Magic Bernoulli's Law Early Kit  | 4–6 yrs   | Early Physics      | Pludo (Robocraze)   | WEB-PLU-032 | Rs. 475    | Soft continuous airflow levitation ball, safe toddler-proof fan nozzle design.   |
| 4 | Little Builders Cable Bridge Project | 4–6 yrs   | Early Engineering  | Pludo (Robocraze)   | WEB-PLU-030 | Rs. 475    | Chunky wooden structural beams, large interlocking joints, bridge load testing.  |
| 5 | 4-in-1 Animals & Dinosaurs Jigsaw    | 3–6 yrs   | Visual Cognition   | Butterfly Edufields | WEB-BUT-010 | Rs. 399    | Thick layered wooden animal anatomy puzzles, fine motor skill coordination.      |

### 4. Primary Vendor Procurement Directory

Direct verified manufacturers and wholesale distributors for bulk request-for-quotation (RFQ).

| Vendor / Supplier   | Total Kits | Priority | Key Kit Lines in Curriculum                                                               | Official Store / Portal                                      |
|---------------------|------------|----------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Pludo (Robocraze)   | 7 Kits     | Tier 1   | Motorized Wind Car, Pneumatic Rocket, SolarGlide, Clap-Racer, Motorized Plane, Robot Claw | <a href="http://robocraze.com">robocraze.com</a>             |
| Mindtronix          | 7 Kits     | Tier 1   | Precision Catapult, Electronic Scale, Cable Car, Forklift, Buji Robot, Arrow Launcher     | <a href="http://mindtronix.in">mindtronix.in</a>             |
| Kiddale             | 8 Kits     | Tier 1   | Paddle Steamer, Glow Lamp, Carousel, Sun-Earth-Moon, T-Rex Walker, Coin Bank, Solar Rover | <a href="http://kiddale.com">kiddale.com</a>                 |
| Butterfly Edufields | 4 Kits     | Tier 1   | Spin Art Machine, 10-in-1 Electricity Lab, Kitchen Chemistry, 150+ Physics/Chem Lab       | <a href="http://butterflyfields.com">butterflyfields.com</a> |
| Practiko            | 3 Kits     | Tier 1   | Wooden Catapult, Secret Cipher Wheel & Maze, Pin Hole Camera & Optical Periscope          | <a href="http://practiko.in">practiko.in</a>                 |
| Kintaro DIY         | 3 Kits     | Tier 2   | Hydraulic Mech Gripper Arm, Harmonic Motion Pendulum, Moto-Bot Bionic Multi-Leg Walker    | <a href="http://kintarodiv.com">kintarodiv.com</a>           |
| BrainHap            | 1 Kit      | Tier 1   | Beginner HD Camera Quadcopter Drone Kit (Demonstration & Pilot Station)                   | <a href="http://brainhap.com">brainhap.com</a>               |